



Sydney Girls High School

2021

ALTERNATE TASK 4

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using a black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11-13, show relevant mathematical reasoning and/or calculations

Total marks:
46

Section I – 10 marks (pages 1-5)

- Attempt Questions 1-10
- Allow about 13 minutes for this section

Section II – 36 marks (pages 6-8)

- Attempt Questions 11-13
- Allow about 47 minutes for this section

THIS IS NOT A TRIAL PAPER

It does not reflect the format or the content of the 2021 HSC Examination Paper in this subject.

Section I

10 marks

Attempt Questions 1–10

Write the question number and your response for Questions 1–10.

1 Which expression is equal to $\int \frac{-1}{\sqrt{1-9x^2}} dx$?

A. $\frac{1}{3} \cos^{-1} 3x + C$

B. $\frac{1}{3} \cos^{-1} \frac{x}{3} + C$

C. $\cos^{-1} 3x + C$

D. $\cos^{-1} \frac{x}{3} + C$

2 Consider the following statement:

‘ n is divisible by 4 and n is not divisible by 8’.

Which of the following is the negation of this statement?

A. n is not divisible by 4 or n is not divisible by 8.

B. n is not divisible by 4 and n is not divisible by 8.

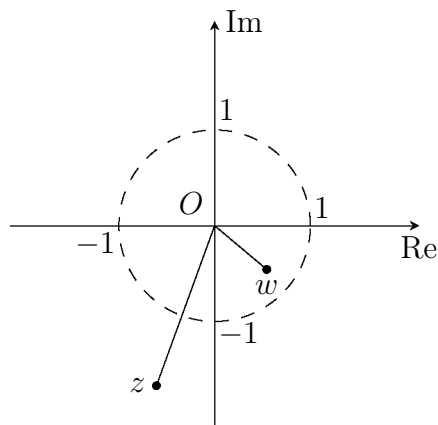
C. n is not divisible by 4 or n is divisible by 8.

D. n is not divisible by 4 and n is divisible by 8.

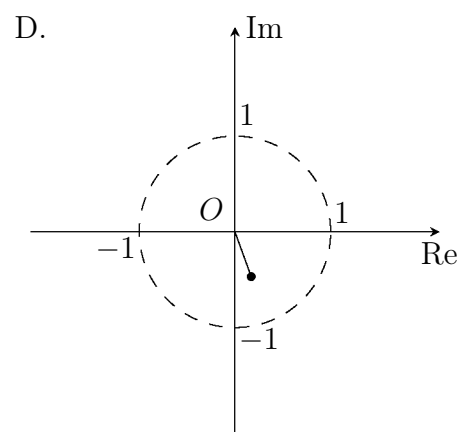
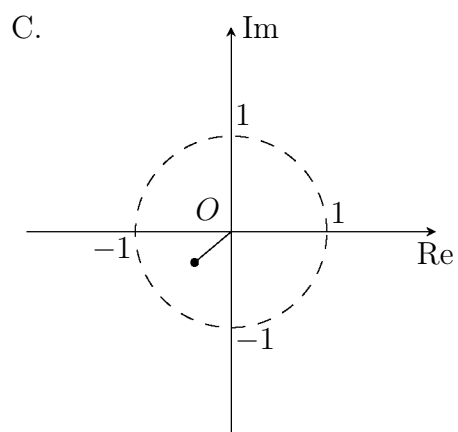
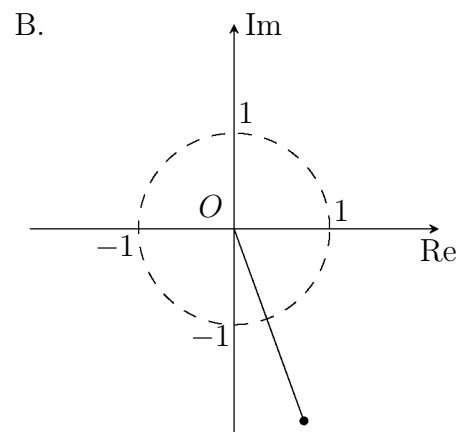
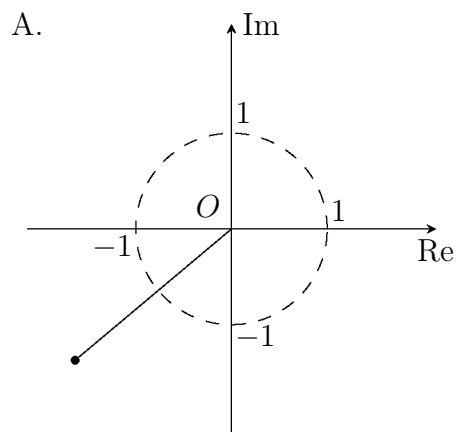
- 3 The acceleration of a particle is given by $\ddot{x} = -\frac{1}{v}$.
What is the displacement of the particle as a function of v ?
- A. $x = \frac{1}{v} + C$
- B. $x = -\frac{v^3}{3} + C$
- C. $x = -\ln|v| + C$
- D. $x = e^{-\frac{1}{2}v^2+C}$
- 4 What is the value of the integral $\int_0^4 \frac{1}{x + \sqrt{x}} dx$ after making the substitution $u = \sqrt{x}$?
- A. $\int_0^{16} \frac{2u}{u^2 + u} du$
- B. $\int_0^{16} \frac{u}{2(u^2 + u)} du$
- C. $\int_0^2 \frac{2u}{u^2 + u} du$
- D. $\int_0^2 \frac{u}{2(u^2 + u)} du$
- 5 When $P(z) = 2z^3 + 3iz^2 + kz - 3i$ is divided by $z^2 + 1$, the remainder is $6z - 6i$.
What is the value of k ?
- A. $k = 5$
- B. $k = 6$
- C. $k = 7$
- D. $k = 8$

- 6 If $\omega \neq 1$ is a cube root of unity, what is $(1 + \omega^2)^3$ equal to?
- A. 1
 - B. ω
 - C. -1
 - D. $-\omega$
- 7 Which of the following is a unit vector in the opposite direction to $-\underline{i} + 2\underline{j} - 2\underline{k}$?
- A. $\frac{1}{3}(\underline{i} - 2\underline{j} + 2\underline{k})$
 - B. $\frac{1}{3}(-\underline{i} + 2\underline{j} - 2\underline{k})$
 - C. $\frac{1}{\sqrt{5}}(\underline{i} - 2\underline{j} + 2\underline{k})$
 - D. $\frac{1}{\sqrt{5}}(-\underline{i} + 2\underline{j} - 2\underline{k})$

- 8 The diagram shows the complex numbers z and w on the Argand diagram. The dotted circle is the unit circle.



Which of the following best shows the position of $\frac{z}{w}$?

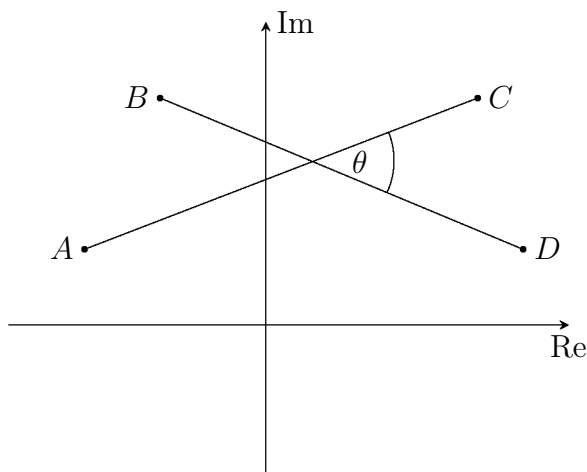


- 9 Consider the following claim:

‘If p is not prime then $(p-1)! + 1$ is not divisible by p ’.

Which of the following is a counterexample to this claim?

- A. A prime number p such that $(p-1)! + 1$ is divisible by p .
 - B. A prime number p such that $(p-1)! + 1$ is not divisible by p .
 - C. A composite number p such that $(p-1)! + 1$ is divisible by p .
 - D. A composite number p such that $(p-1)! + 1$ is not divisible by p .
- 10 The points A , B , C and D on the Argand diagram represent the complex numbers a , b , c and d respectively. The angle θ is the acute angle between AC and BD . Which of the following is a correct expression for θ ?



- A. $\theta = \text{Arg}\left(\frac{a-b}{c-d}\right)$
- B. $\theta = \text{Arg}\left(\frac{c-d}{a-b}\right)$
- C. $\theta = \text{Arg}\left(\frac{c-a}{d-b}\right)$
- D. $\theta = \text{Arg}\left(\frac{d-b}{c-a}\right)$

Section II

36 marks

Attempt Questions 11–13

Start each question on a NEW sheet of paper.

Question 11 (12 marks) Use a NEW sheet of paper.

- (a) Let $z = 1 - \sqrt{3}i$.
- (i) Express z in the form $re^{i\theta}$. **2**
- (ii) Hence find z^{12} in simplest Cartesian form. **1**
- (b) Find $\int \sin^{-1} x \, dx$. **2**
- (c) Find $\int \frac{x^4 + x^2 + 1}{x^2 + 1} \, dx$. **2**
- (d) Recall the standard integral $\int \frac{dx}{\sqrt{x^2 - a^2}} = \ln \left(x + \sqrt{x^2 - a^2} \right) + C$. **2**
- Using this standard integral, find $\int \sqrt{\frac{x+1}{x-1}} \, dx$.
- (e) A particle of mass 5 kg moves in a straight line subject to a force F . **3**
- The particle was initially at rest at the origin.
- If $F = 15 - 10e^{-t}$, determine the displacement of the particle at time t .

End of Question 11

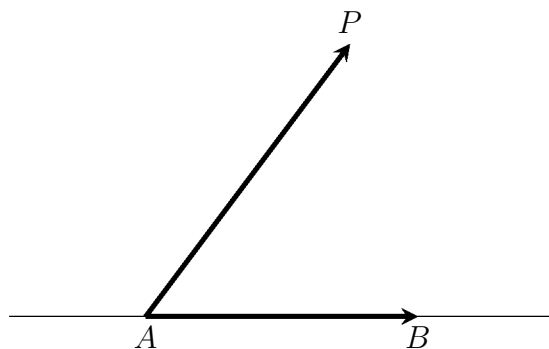
Question 12 (12 marks) Use a NEW sheet of paper.

- (a) (i) Find the square roots of $-18i$. **2**
(ii) Hence, solve $z^2 - (3 + 5i)z - 4 + 12i = 0$. **2**

- (b) Sketch the region on the Argand diagram satisfied by $\operatorname{Re}\left(\frac{1}{z}\right) \leq 1$. **3**

- (c) Consider the points $A(2, -2, 0)$, $B(1, 5, 2)$ and $P(3, 3, 1)$. **2**

Find the projection of $\overrightarrow{AP}$ onto $\overrightarrow{AB}$ in simplest form.



- (d) In the set of real numbers, let P be the proposition:

‘If $a + b$ is irrational then at least one of a and b is irrational.’

- (i) State the contrapositive of the proposition P . **1**
(ii) Write the converse of the proposition P , and state, with reasons, **2**
whether this converse is true or false.

End of Question 12

Question 13 (12 marks) Use a NEW sheet of paper.

(a) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate $\int_0^{\frac{\pi}{3}} \frac{1}{1 + \sin x} dx$. **3**

(b) Consider the recurrence relation defined by $T_1 = 1$ and **3**

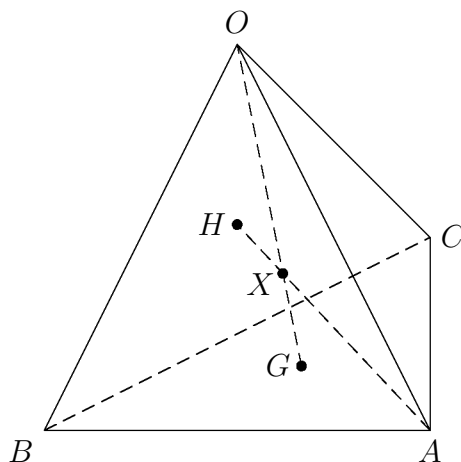
$$T_{n+1} = 3T_n + 4, \text{ for } n = 1, 2, \dots$$

Prove by mathematical induction that

$$T_n = 3^n - 2, \text{ for } n = 1, 2, \dots$$

(c) Prove that if $n \geq 3$ is an odd integer then $n^3 - n$ is divisible by 12. **3**

(d) Let $ABCD$ be a tetrahedron with $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$. **3**



The points G and H satisfy $\overrightarrow{OG} = \frac{\underline{a} + \underline{b} + \underline{c}}{3}$ and $\overrightarrow{OH} = \frac{\underline{b} + \underline{c}}{3}$.

Prove that OG and AH intersect at X , where $\overrightarrow{OX} = \frac{\underline{a} + \underline{b} + \underline{c}}{4}$.

End of task

Multiple choice

Q1	A
Q2	C
Q3	B
Q4	C
Q5	D
Q6	C
Q7	A
Q8	B
Q9	C
Q10	C

Q1

$$\int \frac{-1}{\sqrt{1-9x^2}} dx = \frac{1}{3} \cos^{-1} 3x + C \therefore (A)$$

Q2 $\text{not } (A \text{ and } B) = (\text{not } A) \text{ or } (\text{not } B)$

n is not divisible by 4 OR n is divisible by 8

(C)

Q3

$$x = -\frac{1}{v}$$

$$v \frac{dv}{dx} = -\frac{1}{v}$$

$$\frac{dv}{dx} = -\frac{1}{v^2}$$

$\therefore (B)$

$$\frac{dx}{dv} = -v^2$$

$$x = C - \frac{v^3}{3}$$

Q4 $u = \sqrt{x}$

When $x = 0$, $u = 0$

When $x = 4$, $u = 2$

$$\frac{du}{dx} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}} = \frac{1}{2u}$$

$$\therefore dx = 2u du$$

$$\therefore \int_0^4 \frac{1}{x + \sqrt{x}} dx = \int_0^2 \frac{2u du}{u^2 + u}$$

$$\therefore (C)$$

Q5

$$P(z) = (z^2 + 1)Q(z) + 6z - 6i$$

$$\therefore P(i) = (i^2 + 1)Q(i) + 6i - 6i$$

$$= 0$$

$$\therefore 2i^3 + 3i(i^2) + ki - 3i = 0$$

$$-2 - 3 + k - 3 = 0$$

$$k = 8$$

$$\therefore (D)$$

Q6

Since $\omega \neq 1$ is a cube root of unity

$$\omega^3 = 1 \Rightarrow (\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\Rightarrow 1 + \omega + \omega^2 = 0$$

$$\Rightarrow 1 + \omega^2 = -\omega$$

$$\therefore (1 + \omega^2)^3 = (-\omega)^3$$

$$= -\omega^3$$

$$= -1$$

$$\boxed{\therefore (C)}$$

Q7

$$|-\hat{i} + 2\hat{j} - 2\hat{k}| = \sqrt{(-1)^2 + 2^2 + (-2)^2}$$

$$= \sqrt{1 + 4 + 4}$$

$$= \sqrt{9}$$

$$= 3$$

$\therefore$ Unit vector in opposite direction to $-\hat{i} + 2\hat{j} - 2\hat{k}$

is $-\frac{1}{3}(-\hat{i} + 2\hat{j} - 2\hat{k}) = \frac{1}{3}(\hat{i} - 2\hat{j} + 2\hat{k})$

$$\boxed{\therefore (A)}$$

Q8 Note that $|w| < 1$ and $|z| > 1$.

$$\therefore \left| \frac{z}{w} \right| = \frac{|z|}{|w|} > 1 \quad \therefore \text{Outside unit circle}$$

Also $\arg\left(\frac{z}{w}\right) = \arg(z) - \arg(w)$ with $\arg(z)$ and $\arg(w)$ both negative. Hence

$$\arg(z) < \arg\left(\frac{z}{w}\right) < \arg(w)$$

$$\therefore (B)$$

Q9 If P then Q is equivalent to
If not Q then not P .

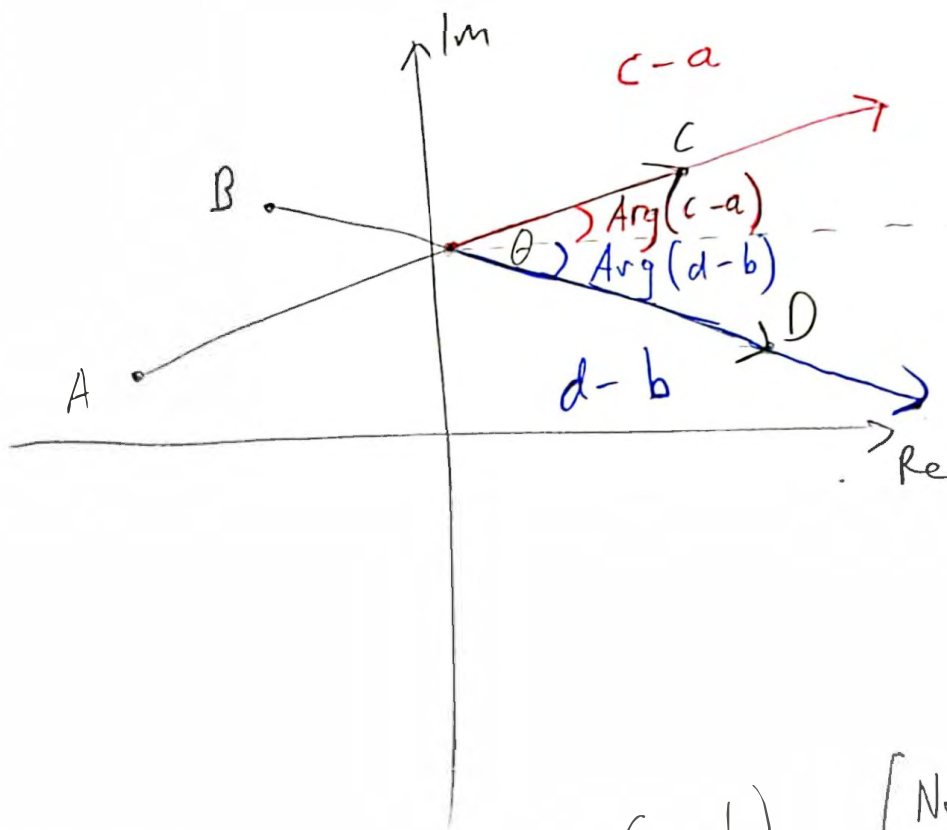
So the claim is equivalent to

"If $(p-1)! + 1$ is divisible by p , then p is prime".

A counter-example is therefore a not prime (composite) number such that $(p-1)! + 1$ is divisible by p .

$$\therefore (C)$$

Q10



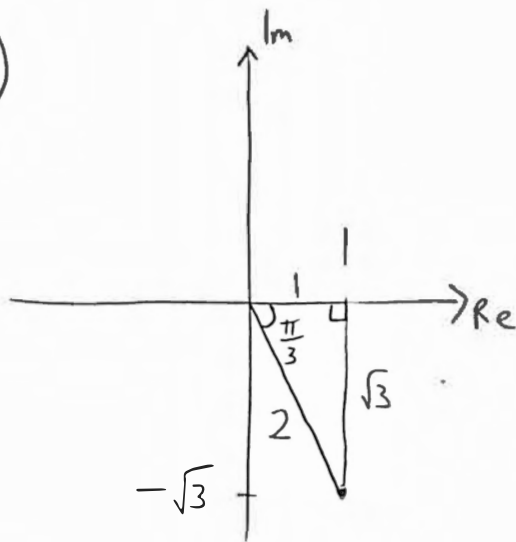
$$\therefore \theta = \text{Arg}(c-a) - \text{Arg}(d-b)$$
$$= \text{Arg}\left(\frac{c-a}{d-b}\right)$$

$\left[\begin{array}{l} \text{Note:} \\ \text{Arg}(d-b) < 0 \end{array} \right]$

$\therefore (c)$

Q11

(a) (i)



$$z = 2e^{-i\pi/3} \checkmark$$

2

- 1 mark for modulus

- 1 mark for argument

$$\begin{aligned} (ii) \quad z^{12} &= \left(2e^{-i\pi/3} \right)^{12} \\ &= 2^{12} e^{-4i\pi} \\ &= 4096 \checkmark \end{aligned}$$

1

$$\begin{aligned} (b) \quad \int \underbrace{\sin^{-1} x}_u \times \underbrace{1}_{v'} dx &= uv - \int u'v dx \\ &= x \sin^{-1} x - \int \frac{1}{\sqrt{1-x^2}} \times x dx \checkmark \\ &= x \sin^{-1} x + \sqrt{1-x^2} + C \checkmark \end{aligned}$$

2

Q11

$$(c) \int \frac{x^4 + x^2 + 1}{x^2 + 1} dx$$

2

$$= \int \frac{x^2(x^2 + 1) + 1}{x^2 + 1} dx$$

$$= \int \left(\frac{x^2(x^2 + 1)}{x^2 + 1} + \frac{1}{x^2 + 1} \right) dx$$

$$= \int \left(x^2 + \frac{1}{x^2 + 1} \right) dx$$

$$= \frac{x^3}{3} + \tan^{-1} x + C$$

$$(d) \int \frac{x}{\sqrt{x^2-1}} dx = \int \frac{x+1}{\sqrt{(x-1)(x+1)}} dx$$

Q 11

$$= \int \left(\frac{x}{\sqrt{x^2-1}} + \frac{1}{\sqrt{x^2-1}} \right) dx \quad \checkmark$$

$$= \sqrt{x^2-1} + \ln(x + \sqrt{x^2-1}) + C \quad \checkmark$$

$$(e) F = ma$$

$$5a = 15 - 10e^{-t} \quad \checkmark$$

$$a = 3 - 2e^{-t}$$

$$v = 3t + 2e^{-t} + C$$

$$\text{When } t=0, v=0 \therefore C = -2$$

$$v = 3t + 2e^{-t} - 2 \quad \checkmark$$

$$x = \frac{3t^2}{2} - 2e^{-t} - 2t + D$$

$$\text{When } t=0, x=0 \therefore D = 2$$

$$x = \frac{3t^2}{2} - 2e^{-t} - 2t + 2 \quad \checkmark$$

Q12

$$(a)(i)(x+iy)^2 = -18i$$

$$(x^2 - y^2) + 2xyi = -18i$$

Equating real and imaginary parts

$$x^2 - y^2 = 0 \quad (1)$$

$$xy = -9 \quad (2)$$

This gives $x+iy = \underline{3-3i}$ and $\underline{-3+3i}$

$$(ii) \Delta = b^2 - 4ac$$

$$= [-(3+5i)]^2 - 4(1)(-4+12i)$$

$$= 9 - 25 + 30i + 16 - 48i$$

$$= -18i$$

$$\therefore z = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$= \frac{3+5i \pm (3-3i)}{2}$$

$$= 3+i, 4i$$

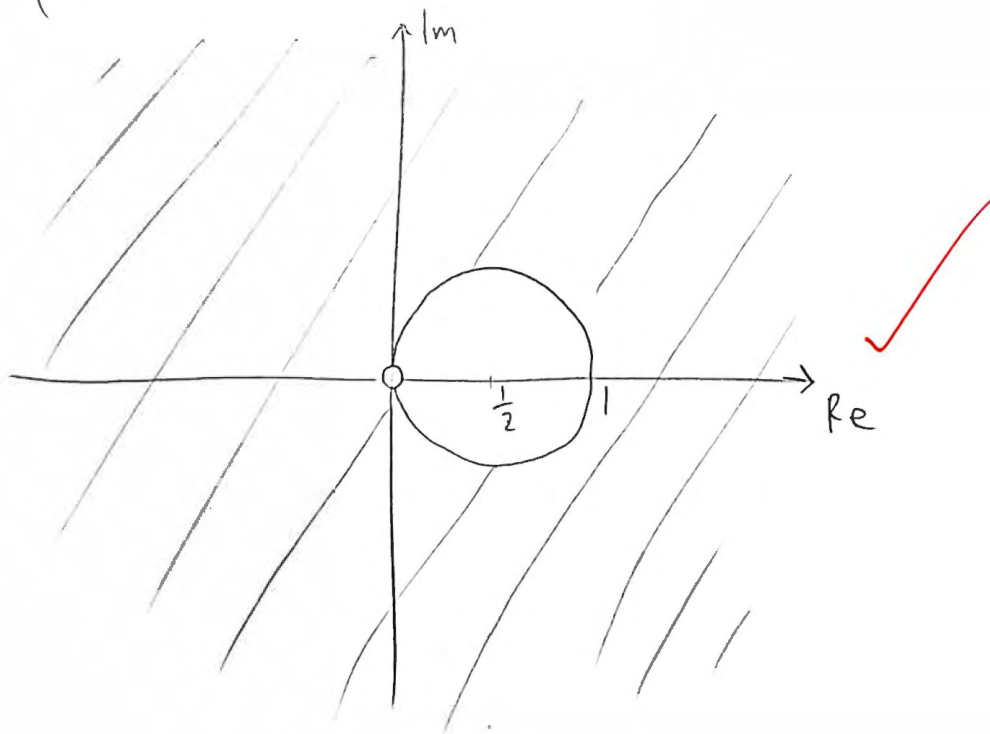
Q12

(b) $z \neq 0$. Let $z = x + iy$

$$\operatorname{Re}\left(\frac{1}{z}\right) = \operatorname{Re}\left(\frac{1}{x+iy} \times \frac{x-iy}{x-iy}\right)$$
$$= \operatorname{Re}\left(\frac{x-iy}{x^2+y^2}\right)$$
$$= \frac{x}{x^2+y^2} \leq 1 \quad \checkmark$$

$$x \leq x^2 + y^2$$

$$x^2 - x + y^2 \geq 0$$
$$\left(x - \frac{1}{2}\right)^2 + y^2 \geq \frac{1}{4} \quad \checkmark$$



- Open circle at the origin
is required for full marks

3

Q 12

$$(c) \vec{AP} = \vec{OP} - \vec{OA} = \begin{pmatrix} 3 \\ 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix}$$

2

$$\vec{AB} = \vec{OB} - \vec{OA} = \begin{pmatrix} 1 \\ 5 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix}$$

$$\text{proj}_{\vec{AB}} \vec{AP} = \left(\frac{\vec{AB} \cdot \vec{AP}}{\vec{AB} \cdot \vec{AB}} \right) \vec{AB}$$

$$= \frac{\begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix}}{\begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix}} \begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix}$$



or equivalent
correct expression
for $\text{proj}_{\vec{AB}} \vec{AP}$

$$= \frac{-1 + 35 + 2}{1 + 49 + 4} \begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix}$$

$$= \frac{36}{54} \begin{pmatrix} -1 \\ 7 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{2}{3} \\ \frac{14}{3} \\ \frac{4}{3} \end{pmatrix}$$



Q12

Contrapositive of P:

(d) (i) If a and b are rational, then $a+b$ is ~~rational~~. ¹ ✓

(ii) Converse of P: ²

If at least one of a and b is irrational, then $a+b$ is irrational. ✓

This converse is false. For a counter-example, consider $a = \sqrt{2}$ and $b = -\sqrt{2}$. Then at least one of a and b is irrational, while $a+b = \sqrt{2} + (-\sqrt{2}) = 0$ is rational. ✓

or equivalent counter-example

e.g. $a = \pi$, $b = -\pi$ e.t.c.

Q15

(a) Let $t = \tan \frac{x}{2}$

When $x = 0$, $t = 0$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

When $x = \frac{\pi}{3}$, $t = \frac{1}{\sqrt{3}}$

$$dt = \frac{1}{2} (t^2 + 1) dx$$

$$dx = \frac{2}{t^2 + 1} dt \quad \checkmark$$

$$\therefore \int_0^{\frac{\pi}{3}} \frac{1}{1 + \sin x} dx = \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1 + \frac{2t}{1+t^2}} \times \frac{2}{t^2 + 1} dt$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{t^2 + 1 + 2t} dt \quad \checkmark$$

$$= 2 \int_0^{\frac{1}{\sqrt{3}}} \frac{1}{(t+1)^2} dt$$

$$= 2 \left[-\frac{1}{t+1} \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= 2 \left(-\frac{1}{\frac{1}{\sqrt{3}} + 1} + 1 \right) \quad \checkmark$$

$$= \sqrt{3} - 1 \quad (\text{after simplifying})$$

simplifying fully not required for full marks

3

(b) ~~Base case:~~ If $n=1$,

Q13

3

$$\begin{aligned} \text{LHS} &= T_1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 3^1 - 2 \\ &= 1 \\ &= \text{LHS} \quad \checkmark \end{aligned}$$

Assume true for $n=k \geq 1$: $T_k = 3^k - 2$

Prove true for $n=k+1$:

R.T.P. $T_{k+1} = 3^{k+1} - 2$

$$\text{LHS} = T_{k+1}$$

$$= 3T_k + 4 \quad \checkmark \quad (\text{by definition})$$

$$= 3(3^k - 2) + 4 \quad (\text{by inductive assumption})$$

$$= 3^{k+1} - 6 + 4$$

$$= 3^{k+1} - 2$$

$$= \text{RHS} \quad \checkmark$$

$\therefore$ True by induction
for $n=1, 2, \dots$

Q13

$$\begin{aligned} \text{(c)} \quad n^3 - n &= n(n^2 - 1) \\ &= n(n-1)(n+1) \\ &= (n-1)n(n+1) \end{aligned}$$

3

Since $n-1$, n and $n+1$ are three consecutive integers, one of them must be divisible by 3. ✓
 $\therefore n^3 - n$ is divisible by 3.

Also, since n is odd we may write $n = 2m+1$, where $m \geq 1$ is an integer.

$$\therefore n^3 - n = [(2m+1)-1](2m+1)[(2m+1)+1] \quad \checkmark$$

sub in
 $n = 2m+1$

$$= 2m(2m+1)(2m+2)$$

$$= 4m(2m+1)(m+1) \quad \checkmark$$

which is divisible by 4.

$\therefore n^3 - n$ is divisible by 12, since it is divisible by 3 and 4.

(d) The line OC_1 is given by:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \vec{OC_1} = \lambda \left(\frac{\underline{a} + \underline{b} + \underline{c}}{3} \right) \quad \lambda \in \mathbb{R}. \quad 3$$

The line AH is given by:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \vec{OA} + \mu \vec{AH} = \underline{a} + \mu \left(\frac{\underline{b} + \underline{c}}{3} - \underline{a} \right), \quad \mu \in \mathbb{R}.$$

These lines intersect when

$$\frac{\lambda}{3} (\underline{a} + \underline{b} + \underline{c}) = (1 - \mu) \underline{a} + \frac{\mu}{3} \underline{b} + \frac{\mu}{3} \underline{c} \quad \checkmark$$

Since $\underline{a}, \underline{b}, \underline{c}$ are linearly independent, we can equate coefficients:

$$\frac{\lambda}{3} = 1 - \mu \quad (1) \quad \text{AND} \quad \frac{\lambda}{3} = \frac{\mu}{3} \quad (2)$$

$$\text{Sub (2) into (1): } \frac{\lambda}{3} = 1 - \lambda \Rightarrow \lambda = \mu = \frac{3}{4}$$

Substitute $\lambda = \frac{3}{4}$ back into the line OC_1 :

$$\begin{aligned} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= \frac{\cancel{3}}{4} \left(\frac{\underline{a} + \underline{b} + \underline{c}}{\cancel{3}} \right) \\ &= \frac{\underline{a} + \underline{b} + \underline{c}}{4}, \quad \checkmark \text{ as required.} \end{aligned}$$